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Department of Electronics Engineering

MINI PROJECT — FINAL YEAR

Submitted on the topic

ARDUINO BASED LPG GAS DETECTOR



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DECLARATION

I, Aman Shukla, a student of Electronics Engineering at the **Government Polytechnic Saharanpur**, hereby declare that the mini project “Arduino Based LPG Gas Detector” is submitted to the Department of Electronics Engineering, **Government Polytechnic Saharanpur**.

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1. ABSTRACT

Liquefied Petroleum Gas (LPG) is a widely used fuel for cooking and heating, but it poses significant safety risks due to its flammability and the potential for leaks. The aim of this project is to design and develop an LPG gas detection system that provides early warnings of gas leaks, ensuring safety in domestic, industrial, and commercial environments. The system is designed to detect the presence of LPG in the air and immediately alert users through visual (LED) and audible (buzzer) signals.

The core of the detection system relies on a highly sensitive gas sensor, such as the MQ-6 or MQ-2, capable of detecting the specific gas concentrations of LPG. The sensor is interfaced with a microcontroller, like Arduino or Raspberry Pi, to continuously monitor the air for gas presence. When the gas level exceeds a preset threshold, the microcontroller activates the alarm system.

Additionally, the project incorporates safety features such as the ability to trigger ventilation systems or shut off gas supply valves upon detecting high gas concentrations. The system also includes a display module (e.g., LCD) to show real-time gas concentration levels for users. By using low-cost components and an intuitive design, this LPG gas detector aims to enhance home and industrial safety, minimizing the risk of fire hazards, explosions, and health issues due to gas leaks.

This project will also focus on the implementation of a user-friendly interface, easy installation, and reliability in continuous operation, making it a feasible and scalable solution for both residential and commercial use. The outcomes of the project will contribute to improving safety measures in environments where LPG is in use and help raise awareness about the importance of gas leak detection.

2. INTRODUCTION

Liquefied Petroleum Gas (LPG) is widely used for cooking, heating, and powering appliances but poses significant safety risks, particularly due to gas leaks. Since LPG is colorless, odorless, and tasteless, detecting leaks can be challenging, especially in large or poorly ventilated areas. If undetected, even small leaks can create explosive environments, leading to fires or explosions. Additionally, prolonged exposure to high concentrations of LPG can cause health issues like respiratory problems and asphyxiation.

This project aims to develop an affordable, reliable LPG gas detection system using an Arduino-based platform. The primary goals are to create a system capable of accurately detecting gas leaks, ensuring early detection and timely alerts.

The system will include an alert mechanism with a buzzer and visual indicators to notify users of potential danger. This solution seeks to enhance safety in homes, businesses, and industrial environments, offering an accessible and effective tool to mitigate LPG-related hazards and improve public safety.

3. BACKGROUND AND LITERATURE REVIEW

Liquefied Petroleum Gas (LPG) is widely used for cooking, heating, and industrial applications, composed mainly of propane and butane. While efficient, LPG poses significant safety risks due to its flammability and odorless nature in its pure form. To aid detection, an odorant like mercaptan is added, but small leaks may still go unnoticed, especially in poorly ventilated spaces. LPG has a low flash point, making it highly combustible and capable of causing explosions or fires if exposed to an ignition source. It is also denser than air, accumulating in low areas, increasing the risk of ignition. High concentrations can lead to asphyxiation, dizziness, or even death. Environmental risks include contributing to greenhouse gases.

Various technologies exist for LPG detection, including electrochemical, semiconductor, and infrared sensors. Electrochemical sensors are accurate but sensitive to environmental changes, while semiconductor sensors are cost-effective but less precise. Infrared sensors offer excellent selectivity and longevity but are more expensive.

Arduino-based systems often utilize semiconductor sensors like MQ-6 or MQ-2, offering a cost-effective and flexible solution for gas leak detection. Arduino's low cost, ease of use, and customization options make it ideal for both domestic and industrial applications, allowing users to build scalable, affordable gas detection systems with alert mechanisms and remote monitoring features.

4. SYSTEM DESIGN AND WORKING PRINCIPLE

4.1 Circuit Design

The circuit design integrates the above-mentioned components to form a functional LPG gas detection system. The main components used in the circuit are as follows:

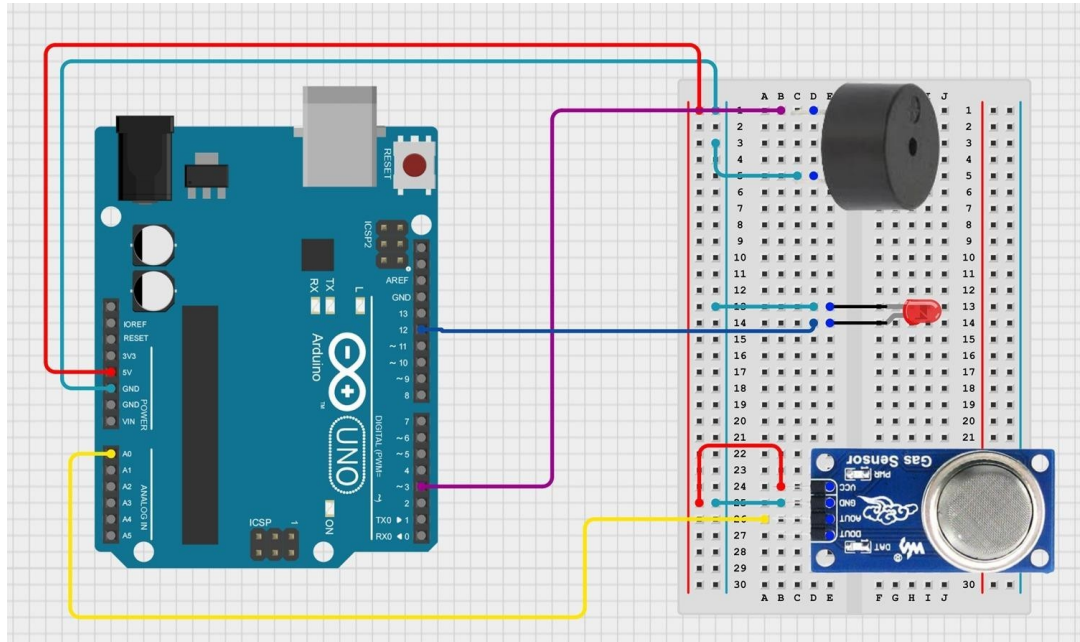


Fig. 4.1: Circuit Diagram of the LPG Gas Detection System

1. Arduino UNO:

- The **Arduino UNO** microcontroller is the central unit of the system. It is responsible for processing the data received from the gas sensor and controlling the outputs such as the buzzer and LEDs. It is an 8-bit microcontroller with multiple input/output pins, which makes it ideal for sensor interfacing and simple control tasks.
- Pins Used:** The analog pin (A0) of the Arduino is used to read the output from the MQ sensor. The digital pins are used to control the buzzer and LEDs.

2. MQ-6 or MQ-2 Gas Sensor:

- The **MQ-6** or **MQ-2** sensor is used for detecting LPG, methane, and other gases. The MQ-6 is specifically sensitive to LPG, while the MQ-2 can detect a wider range of gases including smoke, methane, and butane.
- Operation:** The sensor works on the principle of resistive change. It contains a heating element and a gas-sensitive material. When LPG gas is present in the air, the resistance of the sensor material changes, producing a corresponding voltage output. This signal is sent to the Arduino for analysis.

- **Calibration:** The sensor's output must be calibrated to detect specific gas concentrations effectively. Calibration can be done using known concentrations of LPG.

3. Buzzer (Alarm):

- The **buzzer** provides an audible warning when the LPG concentration exceeds the threshold. The Arduino controls the buzzer by sending a high signal to the buzzer pin, triggering it to emit a sound. The frequency and duration of the sound can be adjusted according to the needs of the system.

4. LEDs (Status Indicators):

- Two LEDs are used to indicate the status of the system. A **green LED** lights up when there is no gas detected or when the system is functioning normally. A **red LED** lights up when the gas concentration exceeds the predefined threshold, signaling a gas leak.
- These LEDs are connected to digital pins of the Arduino and provide visual feedback to users about the operational status of the system.

5. Power Supply:

- The system operates on **5V DC**. This power can be supplied via a **USB cable** (when connected to a computer) or a **5V battery**. The Arduino is powered directly from the power supply, and all other components, such as the gas sensor and the output devices (buzzer and LEDs), are powered through the Arduino board.

Component Connections:

- **MQ Sensor:** The analog output pin of the MQ sensor is connected to an analog input pin of the Arduino (A0). This allows the Arduino to read the gas concentration.
- **Buzzer:** The buzzer is connected to one of the digital output pins (such as pin 8) of the Arduino. When a gas leak is detected, the Arduino outputs a HIGH signal, activating the buzzer.
- **LEDs:** The green and red LEDs are connected to digital pins (such as pins 9 and 10). The Arduino controls these LEDs by sending HIGH or LOW signals to indicate the system status.

4.2 Working Principle

The LPG gas detection system works in the following sequence:

1. Gas Detection:

- The system continuously monitors the environment using the **MQ-6** or **MQ-2 gas sensor**. The sensor detects the presence of LPG in the air by measuring the resistance change in its gas-sensitive material.
- When LPG molecules come into contact with the sensor, the material's resistance decreases, producing a corresponding voltage signal.

2. Signal Processing:

- The sensor's analog output is sent to the **Arduino UNO** microcontroller. The Arduino reads the analog value from the sensor via the analog input pin (A0). This analog signal is then converted into a digital value using the Arduino's **analog-to-digital converter (ADC)**.
- The Arduino program compares the sensor reading against a predefined threshold. The threshold value is set based on the concentration of LPG considered dangerous for the system (typically measured in parts per million, ppm).

3. Threshold Exceedance:

- If the LPG concentration exceeds the threshold, the **Arduino** processes this information and activates the alarm system.
- The Arduino sends a **HIGH signal** to the **buzzer**, causing it to sound and alert the user of a potential gas leak.
- The **red LED** is also turned on to provide a visual indication that a gas leak has been detected.

4. Normal Operation:

- If the LPG concentration is below the threshold, the **green LED** lights up to indicate that the system is functioning normally and no gas leak is present.
- The system continues monitoring the air quality continuously, updating its readings every few seconds.

5. Power Supply:

- The system runs on a **5V DC supply**, ensuring continuous operation of all components. This power can be supplied from a **USB cable** connected to a computer or a **battery pack** for portability.

By implementing this system, users can receive immediate notifications (both audible and visual) in case of a gas leak, minimizing the risk of accidents such as explosions or fires. The system is reliable, cost-effective, and easily deployable, making it ideal for residential, commercial, and industrial applications.

5. COMPONENTS AND MATERIALS

5.1 Arduino Board: Arduino UNO

The Arduino UNO is the central microcontroller used in this LPG gas detection system. It serves as the brain of the system, processing the sensor's input, making decisions based on the readings, and triggering outputs like the buzzer and LEDs.

Specifications of Arduino UNO:

- **Microcontroller:** ATmega328P
- **Operating Voltage:** 5V
- **Input Voltage (recommended):** 7-12V (via DC jack or VIN pin); 5V via USB
- **Digital I/O Pins:** 14 (6 of which can be used as PWM outputs)
- **Analog Input Pins:** 6 (A0-A5), used for reading analog signals from sensors like the MQ series.
- **PWM Pins:** 6 (for simulating analog output using Pulse Width Modulation)
- **Clock Speed:** 16 MHz
- **Flash Memory:** 32 KB (of which 0.5 KB is used by the bootloader)
- **SRAM:** 2 KB
- **EEPROM:** 1 KB
- **Communication:** Serial communication via USB or through hardware (TX/RX pins).
- **Power Supply:** Can be powered by USB (5V) or an external adapter (7-12V).

The Arduino UNO is ideal for this project because of its ease of use, widespread community support, and ample I/O pins to interface with sensors and other output devices like the buzzer and LEDs. It is also compatible with a range of sensors, including the MQ series, making it a versatile choice for gas detection systems.

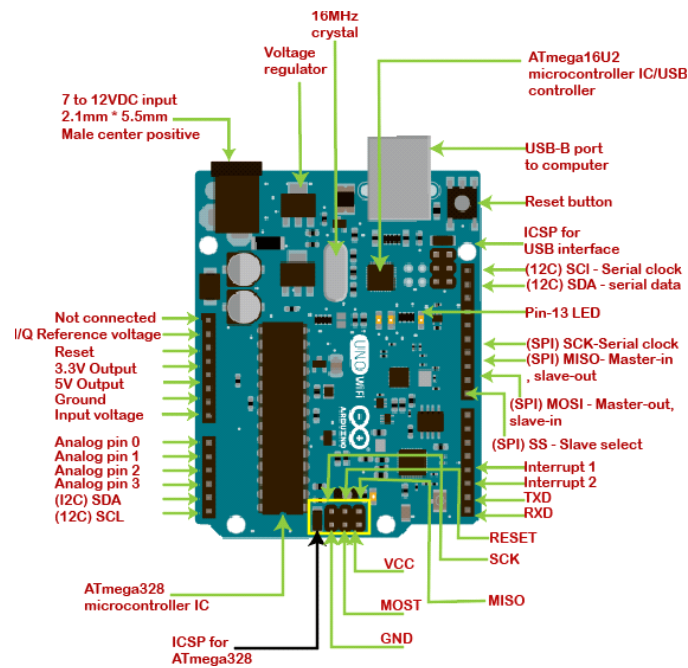


Fig. 5.1: Arduino UNO Board

5.2 Gas Sensor (MQ-6/MQ-2)

The MQ series sensors, including the MQ-6 and MQ-2, are widely used for detecting various gases, including LPG, methane, and smoke. The MQ-6 is specifically designed to detect gases like LPG and butane, while the MQ-2 is more general-purpose and can detect gases like methane, propane, smoke, and alcohol. Both sensors are highly sensitive and reliable for the detection of combustible gases.

Working Principle:

- The MQ sensors operate on the principle of resistive change. They contain a gas-sensitive material (usually SnO_2 or similar compounds) that reacts with gases in the air. When a gas like LPG comes into contact with the material, it alters the material's resistance.
- The heating element inside the sensor warms the sensor's sensitive surface, and when the target gas is present, it causes a chemical reaction that changes the resistance of the surface.
- This change in resistance is then measured by the Arduino through the sensor's analog output. The sensor provides a voltage that correlates with the concentration of the detected gas. The Arduino processes this voltage and determines whether the concentration is above a predefined threshold to activate the alarm.

Why MQ-6 or MQ-2 is Suitable for LPG Detection:

- MQ-6 is specifically designed for LPG and similar gases. It has good sensitivity and can detect small changes in gas concentration.

- The MQ-2 sensor, although more general, is also widely used in detecting gases like propane and methane, making it adaptable for various applications.
- Both sensors are cost-effective, simple to use, and can operate at low voltages, making them an excellent choice for DIY and prototyping projects like this LPG detection system.

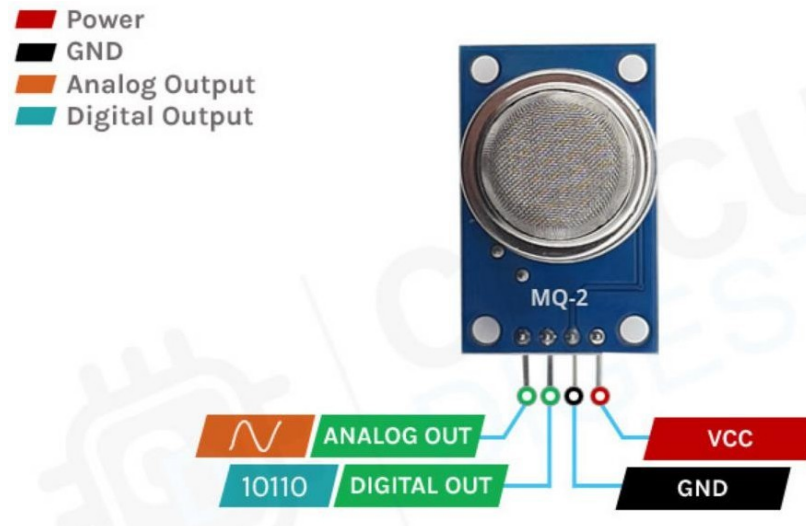


Fig. 5.2: MQ Series Gas Sensor Module

5.3 Buzzer and LEDs

5.3.1 Buzzer

- The buzzer in this system serves as the primary alarm to alert users when a gas leak is detected. A piezoelectric buzzer is typically used in such applications due to its simplicity and reliability. When activated by the Arduino, the buzzer emits a loud sound, notifying users of the detected gas concentration.
- **Working Principle:** The Arduino sends a HIGH signal to the buzzer pin, which causes it to produce sound. The buzzer can be turned off when the gas concentration is below the threshold or when the issue is resolved.
- **Type of Buzzer:** Typically, a passive buzzer is used for its simplicity in generating a tone when powered. Alternatively, an active buzzer can be used to generate sound without requiring additional signal processing.

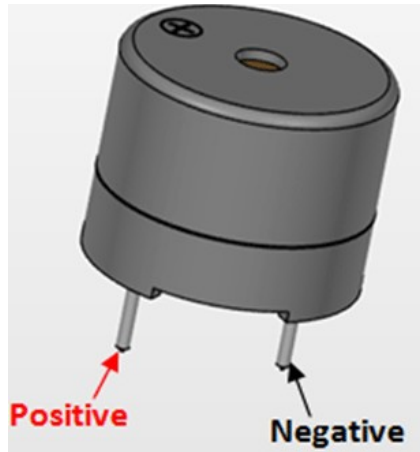


Fig. 5.3: Buzzer

5.3.2 LEDs (Light Emitting Diodes)

- The LEDs provide visual indicators of the system's operational status. Two LEDs are typically used: a green LED and a red LED.
 - **Green LED:** This LED lights up when the gas concentration is normal, indicating that the system is operating without detecting any leaks.
 - **Red LED:** This LED lights up when the gas concentration exceeds the threshold, indicating the presence of a gas leak.
- **Role:** The LEDs are useful for providing a quick visual indication of the system's status. They are especially helpful in noisy environments where audible signals from the buzzer might not be enough.
- **Current-limiting Resistors:** Each LED requires a resistor to limit the current flowing through it and prevent damage. Typical resistor values are in the range of 220Ω to $1k\Omega$, depending on the LED's specifications.

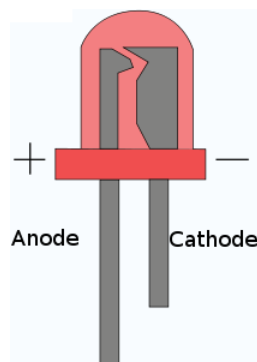


Fig. 5.4: LED (Light Emitting Diode)

5.4 Other Components

In addition to the primary components listed above, the following additional materials are necessary for completing the LPG gas detection system:

1. Wires and Connectors:

- Various jumper wires and connecting wires are needed to make the necessary connections between the components, such as between the Arduino and the gas sensor, buzzer, and LEDs.
- Dupont connectors or male/female headers are commonly used for connecting the sensor and Arduino without soldering.



Fig. 5.5: Jumper Wires

2. Breadboard:

- A breadboard is used to prototype the circuit before soldering the components permanently. It provides a convenient way to make temporary connections and test the circuit before finalizing the design.

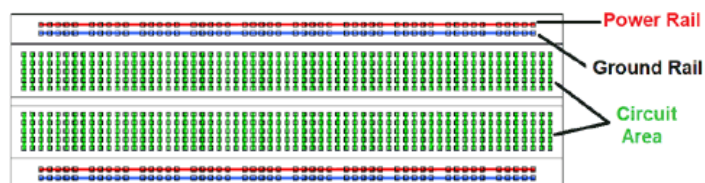


Fig. 5.6: Breadboard

3. Resistors:

- Current-limiting resistors are used for the LEDs to prevent excessive current that could damage them.
- Additional pull-up or pull-down resistors may be needed for certain components, such as buttons or input sensors, though in this project they are not strictly necessary.

4. Power Supply:

- The system can be powered either via a USB connection or a battery pack (e.g., a 5V Li-ion battery or a power adapter). A 5V voltage regulator is used to ensure the system gets a stable voltage, particularly when operating from a battery or external power source.

6. SOFTWARE AND PROGRAMMING

6.1 Arduino IDE and Libraries

The Arduino Integrated Development Environment (IDE) is the primary software platform used to write, compile, and upload the code to the Arduino board. The Arduino IDE is open-source and provides a user-friendly interface for writing programs (called sketches) in C/C++ for microcontroller-based projects. It is designed for ease of use and is well-suited for beginners and experienced programmers alike.

The IDE allows users to:

1. Write code using a simple text editor.
2. Verify and compile the code to check for errors.
3. Upload the compiled code to the Arduino board.
4. Use the Serial Monitor to display outputs for debugging or monitoring real-time data.

For the LPG gas detection project, the Arduino IDE is used to program the Arduino board and interface with the MQ-6 or MQ-2 gas sensor, buzzer, and LEDs.

Libraries: In some cases, external libraries are required to facilitate sensor reading and simplify the code. While the MQ sensors (MQ-2, MQ-6) typically don't require specific libraries, users may need additional libraries for tasks such as serial communication, or when using advanced sensors. However, for this basic LPG detection system, the built-in `analogRead()` function in the Arduino IDE is sufficient to read the data from the gas sensor.

Library Requirements:

- For serial communication, Arduino provides a built-in library, `Serial`, which allows data to be sent and received over the serial port.
- Optional libraries for tasks like temperature or humidity readings, if the system is expanded in the future, are readily available from the Arduino Library Manager.

6.2 Code Explanation

The core of the LPG gas detection system is the code that runs on the Arduino. Below is the general outline of the code for this project, along with a breakdown of its sections:

6.2.1 Initializing the Sensor

The first step in the code is to initialize the sensor, which involves configuring the analog pin that will read the sensor's output. The MQ-6 or MQ-2 sensor sends an analog voltage to the Arduino proportional to the gas concentration, and this value is read by the Arduino's analog input pins.

```
// Define pin numbers
int gasSensorPin = A0; // Analog pin to which MQ sensor is
                        // connected
int buzzerPin = 8;      // Digital pin for buzzer
int greenLEDPin = 9;    // Digital pin for Green LED
int redLEDPin = 10;     // Digital pin for Red LED

// Threshold value for gas concentration
int threshold = 300;    // Set the threshold for gas detection
                        // (this value depends on calibration)
```

In the above section:

- gasSensorPin is connected to the output of the MQ sensor (analog output pin).
- buzzerPin, greenLEDPin and redLEDPin correspond to the Arduino pins controlling the buzzer and LEDs.
- threshold is the predefined value for gas concentration above which the alarm should be triggered.

6.2.2 Reading Sensor Data

To read the sensor data, the Arduino uses the `analogRead()` function. This function returns a value between 0 and 1023 based on the input voltage (from the sensor).

```
void setup() {
  // Initialize the pins
  pinMode(buzzerPin, OUTPUT);
  pinMode(greenLEDPin, OUTPUT);
  pinMode(redLEDPin, OUTPUT);

  // Start serial communication for debugging purposes
  Serial.begin(9600);
}
```

- `setup()` is the function that runs once when the Arduino is powered on or reset. In this function:
 - The pins for the buzzer and LEDs are set as output.
 - The serial communication is initialized to allow for debugging and monitoring data via the Serial Monitor.

6.2.3 Setting Thresholds for Gas Concentration

In the main loop of the program, the gas concentration is read from the sensor. The system compares this value to the threshold, and based on the comparison, it triggers the buzzer and LEDs.

```
void loop() {  
  // Read sensor value (analog voltage)  
  int sensorValue = analogRead(gasSensorPin);  
  
  // Print the sensor value to the Serial Monitor for debugging  
  Serial.println(sensorValue);  
  
  // Check if gas concentration exceeds threshold  
  if (sensorValue > threshold) {  
    // Gas detected above threshold  
    digitalWrite(buzzerPin, HIGH);    // Activate the buzzer  
    digitalWrite(redLEDPin, HIGH);    // Turn on Red LED  
    digitalWrite(greenLEDPin, LOW);   // Turn off Green LED  
  } else {  
    // No gas detected or gas concentration is safe  
    digitalWrite(buzzerPin, LOW);     // Deactivate the buzzer  
    digitalWrite(redLEDPin, LOW);     // Turn off Red LED  
    digitalWrite(greenLEDPin, HIGH);  // Turn on Green LED  
  }  
  
  delay(1000); // Delay for a short period before the next  
  reading  
}
```

In the above section:

- The sensorValue is read from the MQ sensor. This value is printed to the Serial Monitor for real-time monitoring of the gas concentration.
- The program compares the sensor value with the threshold. If the value exceeds the threshold, the buzzer is activated, and the red LED lights up. Simultaneously, the green LED is turned off to indicate a gas leak.
- If the concentration is below the threshold, the buzzer is turned off, and the green LED is lit, indicating safe levels.

6.2.4 Activating the Alarm When Gas Concentration Exceeds the Threshold

When the sensor detects that the gas concentration is above the threshold, the system triggers an alarm (buzzer) and a visual alert (red LED) to notify the user. The buzzer and red LED remain active until the gas concentration returns to safe levels.

```
if (sensorValue > threshold) {  
    digitalWrite(buzzerPin, HIGH);    // Activate the buzzer  
    digitalWrite(redLEDPin, HIGH);    // Turn on Red LED  
    digitalWrite(greenLEDPin, LOW);   // Turn off Green LED  
} else {  
    digitalWrite(buzzerPin, LOW);     // Deactivate the buzzer  
    digitalWrite(redLEDPin, LOW);     // Turn off Red LED  
    digitalWrite(greenLEDPin, HIGH);  // Turn on Green LED  
}
```

This block of code ensures that the system reacts immediately when a gas leak is detected and alerts the user through both auditory and visual cues.

6.2.5 Using LEDs for Status Indication

The LEDs serve as visual indicators. When the gas concentration is normal, the green LED lights up to indicate that everything is functioning properly. When a gas leak is detected, the red LED lights up, signaling an unsafe condition.

```
digitalWrite(redLEDPin, LOW);    // Turns off the red LED when  
gas is not detected  
digitalWrite(greenLEDPin, HIGH); // Turns on the green LED when  
gas is not detected
```

These actions occur every time the sensor data is read and compared against the threshold.

6.2.6 Optional: Serial Monitor for Debugging or Displaying Values

For debugging and monitoring purposes, the sensor values are sent to the Serial Monitor. This helps in troubleshooting and allows the user to see the actual sensor readings.

```
Serial.println(sensorValue); // Output sensor value to Serial  
Monitor
```

By using the Serial Monitor, users can observe the gas concentration in real-time, which helps in calibrating the sensor or checking the system's functionality.

6.3 Flowchart

A flowchart can visually represent the logic behind the software and the decision-making process. Below is a simple flowchart explaining the process of detecting gas levels, processing data, and triggering the alarm:

Flowchart Logic:

1. **Start**
 2. **Initialize Sensor:** Set up sensor, buzzer, and LEDs.
 3. **Read Sensor Data:** Obtain gas concentration from the MQ sensor.
 4. **Check Threshold:** Is the gas concentration above the threshold?
 - If YES: Trigger buzzer and red LED, turn off green LED.
 - If NO: Turn off buzzer and red LED, turn on green LED.
- Repeat:** Go back to reading sensor data.

7. TESTING AND IMPLEMENTATION

Liquefied Petroleum Gas (LPG), primarily composed of propane and butane, is commonly used in cooking, heating, and industrial applications. While LPG is efficient and convenient, it poses significant safety risks due to its flammability and odorless nature in its pure form.

To help detect leaks, an odorant like mercaptan is added, but small leaks can still go unnoticed, especially in poorly ventilated areas. LPG's low flash point makes it highly combustible, capable of causing explosions or fires if exposed to an ignition source. Additionally, LPG is denser than air, causing it to accumulate in low-lying areas, which increases the risk of ignition. High concentrations of LPG can also lead to asphyxiation, dizziness, or even death. Beyond these immediate dangers, LPG contributes to environmental risks, including greenhouse gas emissions.

Various sensor technologies exist for detecting LPG, such as electrochemical, semiconductor, and infrared sensors. Electrochemical sensors offer high accuracy but are sensitive to environmental changes. Semiconductor sensors, like those used in Arduino-based systems, are cost-effective and easy to implement, though they are less precise. Infrared sensors are more selective and durable, but their higher cost limits their widespread use.

Arduino-based gas detection systems frequently utilize semiconductor sensors like the MQ-6 or MQ-2. These sensors provide a flexible, affordable solution for detecting LPG leaks, making them suitable for both domestic and industrial applications. The popularity of Arduino lies in its low cost, ease of use, and customization capabilities, which allow users to create scalable systems. Such systems can include alert mechanisms (e.g., buzzers, LEDs) and features for remote monitoring, ensuring timely detection and response to gas leaks. This makes Arduino-based systems an effective choice for enhancing safety and reducing risks associated with LPG.

8. RESULTS AND DISCUSSION

The performance of the Arduino-based LPG gas detection system was evaluated based on several parameters, including sensor accuracy, response time, and alarm effectiveness. The MQ-6 sensor demonstrated reasonable accuracy, though environmental factors such as temperature and humidity influenced its performance. It provided a qualitative indication of LPG presence, triggering the alarm when the gas concentration exceeded the set threshold. The system responded within 5-10 seconds, which is acceptable for safety applications, and successfully used a buzzer and LED for timely warnings.

When compared to commercial LPG detectors, the Arduino-based system was more cost-effective but lacked the precision and advanced features of higher-end models. Commercial systems typically offer more accurate sensors (e.g., electrochemical or infrared), faster response times, and additional features like real-time displays and connectivity. However, the Arduino system is ideal for budget-conscious users or DIY projects.

Despite its advantages, the Arduino-based system has limitations, including the MQ-6 sensor's sensitivity to environmental factors and the need for calibration. It also lacks features like battery backup or digital displays, and its sensor has a limited lifespan. While suitable for basic applications, commercial systems are more reliable and durable for critical or long-term use.

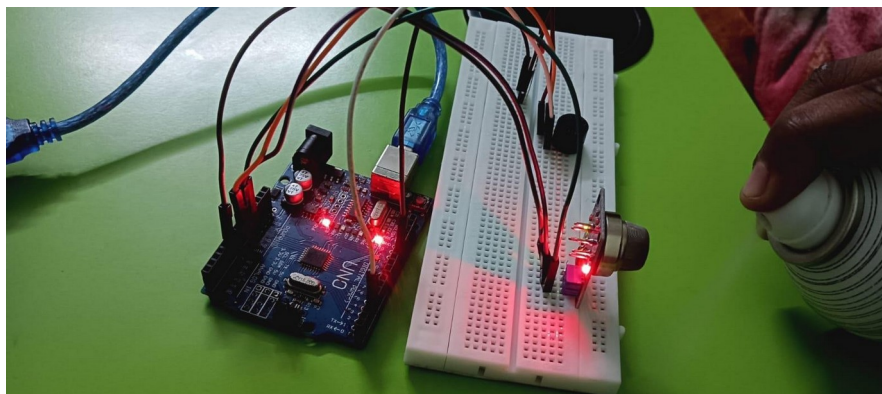


Fig. 8.1: Prototype of the LPG Gas Detector During Testing

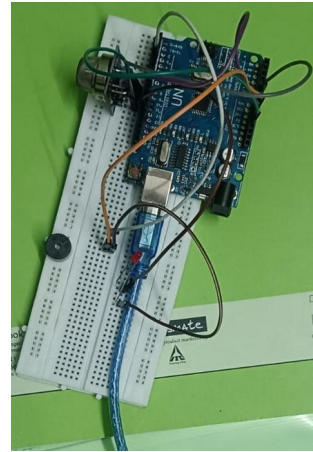
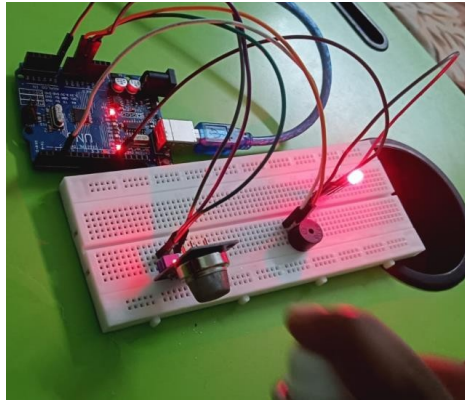


Fig. 8.2: Prototype Setup